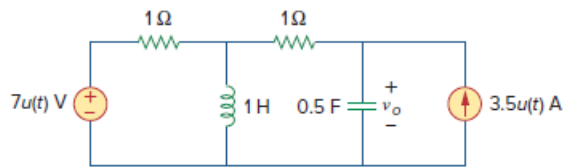


Problem

Find $v_o(t)$, for all $t > 0$, in the circuit of Fig. 16.53.

Figure 16.53:



Step-by-step solution

Step 1 of 7

Refer to circuit diagram in Figure 16.53 in the textbook.

It is known that,

$$u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

For $t < 0$ the capacitor acts like an open circuit and inductor acts like a short circuit.

The initial conditions for the circuit at $t < 0$ is,

$$i_L(0) = 0 \text{ A}$$

$$v_o(0) = 0 \text{ V}$$

Step 2 of 7

The s-domain circuit is shown in Figure 1.

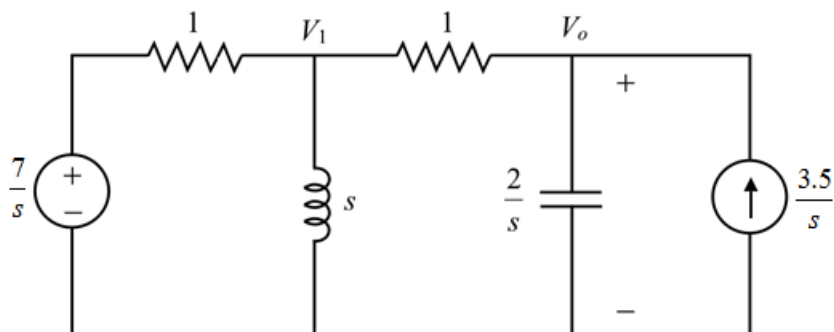


Figure 1

Step 3 of 7

Apply the Kirchhoff's current law at node V_1 .

$$\frac{V_1 - \frac{7}{s}}{1} + \frac{V_1}{s} + \frac{V_1 - V_o}{1} = 0$$

$$\left(1 + \frac{1}{s} + 1\right)V_1 = \frac{7}{s} + V_o$$

$$\left(\frac{2s+1}{s}\right)V_1 = \frac{7+sV_o}{s}$$

$$V_1 = \frac{7+sV_o}{2s+1} \dots\dots (1)$$

Step 4 of 7

Apply the Kirchhoff's current law at node V_o .

$$\frac{V_o - V_1}{1} + \frac{V_o}{\frac{2}{s}} - \frac{3.5}{s} = 0$$

$$\left(1 + \frac{s}{2}\right)V_o - V_1 = \frac{3.5}{s} \quad \text{Substitute } \frac{7+sV_o}{2s+1} \text{ for } V_1 \text{ in the above equation.}$$

$$\left(\frac{s+2}{2}\right)V_o - V_1 = \frac{3.5}{s}$$

$$\left(\frac{s+2}{2}\right)V_o - \frac{7+sV_o}{2s+1} = \frac{3.5}{s}$$

$$\left(\frac{s+2}{2} - \frac{s}{2s+1}\right)V_o = \frac{3.5}{s} + \frac{7}{2s+1}$$

$$\left(\frac{(s+2)(2s+1) - 2s}{2(2s+1)}\right)V_o = \frac{3.5(2s+1) + 7s}{s(2s+1)}$$

$$\left(\frac{2s^2 + 5s + 2 - 2s}{2}\right)V_o = \frac{7s + 3.5 + 7s}{s}$$

$$V_o = \frac{14s + 3.5}{s(s^2 + 1.5s + 1)} \dots\dots (2)$$

Step 5 of 7

Simplify the equation (2) using partial fractions.

$$V_o = \frac{A}{s} + \frac{Bs+C}{s^2 + 1.5s + 1} \dots\dots (3)$$

$$14s + 3.5 = A(s^2 + 1.5s + 1) + s(Bs + C)$$

$$14s + 3.5 = (A+B)s^2 + (1.5A+C)s + A \dots\dots (4)$$

Equate the constant terms from equation (4) on both sides.

$$A = 3.5$$

Equate the s^2 terms from equation (4) on both sides.

$$A + B = 0$$

$$B = -A$$

$$B = -3.5$$

Equate the s terms from equation (4) on both sides.

$$1.5A + C = 14$$

$$(1.5)(3.5) + C = 14$$

$$C = 8.75$$

Step 6 of 7

Recall equation (3).

$$V_o = \frac{A}{s} + \frac{Bs+C}{s^2+1.5s+1}$$

Substitute 3.5 for A , -3.5 for B , and 8.75 for C in the above equation.

$$\begin{aligned} V_o &= \frac{3.5}{s} + \frac{-3.5s+8.75}{s^2+1.5s+1} \\ &= \frac{3.5}{s} + \frac{-3.5\left(s+\frac{3}{4}\right)+11.375}{\left(s+\frac{3}{4}\right)^2+\left(\frac{\sqrt{7}}{4}\right)^2} \\ &= \frac{3.5}{s} - \frac{3.5\left(s+\frac{3}{4}\right)}{\left(s+\frac{3}{4}\right)^2+\left(\frac{\sqrt{7}}{4}\right)^2} + \frac{(17.2)\left(\frac{\sqrt{7}}{4}\right)}{\left(s+\frac{3}{4}\right)^2+\left(\frac{\sqrt{7}}{4}\right)^2} \end{aligned}$$

Step 7 of 7

It is known that,

$$\mathcal{L}^{-1}\left[\frac{1}{s}\right] = u(t)$$

$$\mathcal{L}^{-1}\left[\frac{s+a}{(s+a)^2+\omega^2}\right] = e^{-at} \cos \omega t$$

$$\mathcal{L}^{-1}\left[\frac{\omega}{(s+a)^2+\omega^2}\right] = e^{-at} \sin \omega t$$

Apply inverse Laplace transform.

$$\begin{aligned} v_o(t) &= 3.5u(t) - 3.5e^{-\frac{3}{4}t} \cos\left(\frac{\sqrt{7}}{4}t\right)u(t) + 17.2e^{-\frac{3}{4}t} \sin\left(\frac{\sqrt{7}}{4}t\right)u(t) \\ &= \left[3.5 - 3.5e^{-\frac{3}{4}t} \cos\left(\frac{\sqrt{7}}{4}t\right) + 17.2e^{-\frac{3}{4}t} \sin\left(\frac{\sqrt{7}}{4}t\right)\right]u(t) \end{aligned}$$

Therefore, the output voltage, $v_o(t)$ for all $t > 0$ is,

$$\boxed{\left[3.5 - 3.5e^{-\frac{3}{4}t} \cos\left(\frac{\sqrt{7}}{4}t\right) + 17.2e^{-\frac{3}{4}t} \sin\left(\frac{\sqrt{7}}{4}t\right)\right]u(t)}.$$